

# CSE4502 (Operating Systems Lab 9): 2B

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February 26, 2026

## 1 System Requirements

You must develop a simulation that accepts user input to define the environment and the execution behavior.

### 1.1 User Inputs

- Two integers:  $N$   $P$  where  $N$  is the number of philosophers and  $P$  is the phase which can be either 1 or 2.

## 2 Phase 1: Deadlock Demonstration

When this phase is selected, the system must demonstrate a state of **Deadlock**.

- This Phase demonstrates deadlock, the code is stuck in a loop and does not end its execution.

## 3 Phase 2: Resolved Simulation

In this phase, the Deadlock is not present and the code execution terminates.

## 4 Visualization & Output

- **Real-time State Tracking:** Display whether each philosopher is currently waiting for 2 forks or waiting for 1 or eating.
- **Termination:** Prompt when the system terminates.